A1.

SentinelX Cloud Threat Intelligence Using NLP and Serverless Risk Scoring

A2 Main User Action:

Users submit a suspicious URL to the cloud for the threat Analysis.

Exact Request:

"Scan This suspicious URL Calculate its threat Score, store the report in the cloud, and display whether it is safe or malicious."

PART - B - Network Journey

B1. Message Path

Android APP / web Browser
↓
DNS server
↓
HTTPS / TLS
↓
TCP/IP
↓
Wi-fi / 4G / 5G
↓
Router
↓
ISP

ISP
↓
Internet
↓
Cloud API Gateway
↓
Spring Boot Server
↓
Authentication
↓
Threat Detech service
↓
AI / NLP Model
↓
firestore DB
↓
Response to use

Big "Technical Translation" Table

Layer/Component	What it does in your app request	Possible failure	Cloud-based protection
DNS	Resolves SentinelX domain name into server IP address	DNS lookup failure	DNS redundancy
HTTPS/TLS	Encrypts communication b/w user and cloud	Invalid certificate	SSL/TLS certificate validation
TCP/IP	Transfers packets reliably between devices & server	Packet loss	Retransmission & acknowledgement
Router/ISP/Internet	Routes packet to the cloud server	Network congestion	Multiple routing paths & cloud availability zones
API Gateway/Load Balancer	Recieves requests and distributes them among backend servers	High traffic	Auto Load Balancer and Auto Scalling

[illegible] Cloud Landing Zone

Cloud Need	Component Choice	Service Model	Justification
User Login	Firebase Authentication	PaaS	Secure user authentication without managing servers
Backend Business Logic	Spring Boot REST API	PaaS	Executes URL scanning and AI processing
Database	Firebase Firestore	PaaS	Stores users, scan history and reports
File Storage	Firebase Storage	PaaS	Stores uploaded APKs and screenshots
Monitoring	Firebase Analytics + Cloud Monitoring	SaaS	Monitors requests, errors and application performance

PART D – API Contract

Field	Answer
Endpoint URL	/api/v1/url/scan
HTTP Method	POST
Request Data	User ID, URL, Device Type, Timestamp
Response Data	Threat Score, Prediction (Safe/Malicious), Risk level, Recommendation
Authentication Rule	Firebase JWT Token Required
Error case	Invalid URL, Unauthorized User (401), Server Error (500), AI prediction failure

Part E – Elasticity, Scalability and Measured Service

E1. 7200 Requests in 10 Minutes

Peak request/minute

7200/10 = 720 request/minute.

E2. 25% Spare Capacity.

720 * 1.25 = 900 requests/minute

E3 One instance handles : 180 request/minute

Required Instance : 900/180 = 5 Cloud Instances

E4. Elastic / Scalable Decision

Configure Spring Boot API servers with automatic scaling based on CPU utilization and request count. A load balancer distributes incoming traffic evenly among multiple cloud instances during peak loads.

E5. Measured Service Metrics

1. API Request Count per Minute
2. Average Response Time (Latency)

Part - E - Google Research Evidence

Search / prompt used : "How does firebase Authentication secure users in cloud application?"

Tools used : Google / ChatGpt

Useful findings : firebase Authentication provides secure user login supports email/password and google sign-in, and issues authentication token to verify users before accessing cloud resources

How I applied it : I used firebase Authentication to verify every user before allowing access to sentinelX threat scanning service